
Healthcheck/LivenessProbe in Kubernetes

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Manifest to create a POD along with the LivenessProbe

```
root@ip-172-31-33-162: /home/ubuntu
apiVersion: v1
kind: Pod
metadata:
  labels:
    test: liveness
  name: mylivenessprobe
spec:
  containers:
    - name: liveness
      image: ubuntu
      args:
        - /bin/sh
        - -c
        - "touch /tmp/healthy; sleep 1000"
      livenessProbe:
        exec:
          command:
            - cat
            - /tmp/healthy
        initialDelaySeconds: 5
        periodSeconds: 5
        timeoutSeconds: 30
```

-- INSERT --

Activate Windows
Go to Settings to activate Windows.

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1.

(This Kubernetes manifest defines a Pod named mylivenessprobe with a container named liveness running the ubuntu image. The container runs a simple shell command (touch /tmp/healthy; sleep 1000), which creates a file /tmp/healthy and then sleeps for 1000 seconds. The Pod includes a liveness probe that uses an exec check to run the command cat /tmp/healthy every 5 seconds, starting 5 seconds after the container starts (initialDelaySeconds: 5). If the file /tmp/healthy exists, the probe will succeed. If the file is missing or the command takes more than 30 seconds to execute, the probe will fail, and Kubernetes may restart the container. This setup helps ensure the container is healthy by checking if the file is accessible, demonstrating the functionality of liveness probes in Kubernetes.)

```
apiVersion: v1
kind: Pod
metadata:
labels:
test: liveness
name: mylivenessprobe
spec:
containers:
- name: liveness
image: ubuntu
args:
- /bin/sh
- -c
- "touch /tmp/healthy; sleep 1000"
livenessProbe:
exec:
command:
- cat
- /tmp/healthy
initialDelaySeconds: 5
periodSeconds: 5
timeoutSeconds: 30
```

Apply the manifest..Enter the container and run a Health Check

```
root@mylivenessprobe: /
root@ip-172-31-33-162:/home/ubuntu# vi liveness.yml
root@ip-172-31-33-162:/home/ubuntu# kubectl apply -f liveness.yml
pod/mylivenessprobe created
root@ip-172-31-33-162:/home/ubuntu# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
mylivenessprobe                    1/1     Running   0           12s
root@ip-172-31-33-162:/home/ubuntu# kubectl exec mylivenessprobe -it -- /bin/bash
root@mylivenessprobe:/# echo $?
0
root@mylivenessprobe:/#
```

2.

commands:

echo \$?

Zero Exit Code (0):

- **Indicates Success:** If the command executed by the liveness probe returns an exit code of 0, it means the probe has succeeded. Kubernetes interprets this as the container being in a healthy state.
- **Example:** Running `cat /tmp/healthy` will return 0 if the file exists and is readable.

Non-Zero Exit Code (>0):

- **Indicates Failure:** If the command executed by the liveness probe returns a non-zero exit code, it means the probe has failed. Kubernetes interprets this as the container being unhealthy.
- **Response:** Kubernetes will take the configured action for a failed liveness probe (typically restarting the container).
- **Example:** If `cat /tmp/healthy` is run and the file does not exist, the command will return an error and a non-zero exit code.

This liveness probe is using an exec check to run the command cat /tmp/healthy every 5 seconds

```
root@ip-172-31-33-162:/home/ubuntu# vi liveness.yml
root@ip-172-31-33-162:/home/ubuntu# kubectl apply -f liveness.yml
pod/mylivenessprobe created
root@ip-172-31-33-162:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
mylivenessprobe 1/1     Running   0          12s
root@ip-172-31-33-162:/home/ubuntu# kubectl exec mylivenessprobe -it -- /bin/bash
root@mylivenessprobe:/# echo $?
0
root@mylivenessprobe:/# ls /tmp/healthy
/tmp/healthy
root@mylivenessprobe:/#
```

3.

The file is present in the container so everything is okay.

Lets Run a command which is bound to fail to see what exit code we get.

```
root@ip-172-31-33-162:/home/ubuntu# vi liveness.yml
root@ip-172-31-33-162:/home/ubuntu# kubectl apply -f liveness.yml
pod/mylivenessprobe created
root@ip-172-31-33-162:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
mylivenessprobe 1/1     Running   0          12s
root@ip-172-31-33-162:/home/ubuntu# kubectl exec mylivenessprobe -it -- /bin/bash
root@mylivenessprobe:/# echo $?
0
root@mylivenessprobe:/# ls /tmp/healthy
/tmp/healthy
root@mylivenessprobe:/# ls rohit
ls: cannot access 'rohit': No such file or directory
root@mylivenessprobe:/# echo $?
2
root@mylivenessprobe:/#
```

4.

We ran “ls rohit” command which generated an **Error**
And we are getting a **Non-Zero Exit Code (>0)** i.e “2”

Lets DELETE the key health check file and see What happens.

```

root@ip-172-31-33-162:/home/ubuntu# vi liveness.yml
root@ip-172-31-33-162:/home/ubuntu# kubectl apply -f liveness.yml
pod/mylivenessprobe created
root@ip-172-31-33-162:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
mylivenessprobe 1/1     Running   0           12s
root@ip-172-31-33-162:/home/ubuntu# kubectl exec mylivenessprobe -it -- /bin/bash
root@mylivenessprobe:/# echo $?
0
root@mylivenessprobe:/# ls /tmp/healthy
/tmp/healthy
root@mylivenessprobe:/# ls rohit
ls: cannot access 'rohit': No such file or directory
root@mylivenessprobe:/# echo $?
2
root@mylivenessprobe:/# cd tmp
root@mylivenessprobe:/tmp#
root@mylivenessprobe:/tmp# healthy
root@mylivenessprobe:/tmp# rm healthy
root@mylivenessprobe:/tmp# ls
root@mylivenessprobe:/tmp# echo $?
0
root@mylivenessprobe:/tmp# echo $?
B
root@mylivenessprobe:/tmp# command terminated with exit code 137
root@ip-172-31-33-162:/home/ubuntu#

```

5.

Health check/LivenessProbe Failed

THE POD GOT TERMINATED WITH EXIT CODE 137 AND WE GOT SHIFTED BACK TO THE NODE!

Old POD was killed New POD got created because of LivenessProbe failure.

```

0
root@mylivenessprobe:/tmp# echo $?
0
root@mylivenessprobe:/tmp# command terminated with exit code 137
root@ip-172-31-33-162:/home/ubuntu# kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
mylivenessprobe 1/1     Running   1 (23s ago)  3m53s
root@ip-172-31-33-162:/home/ubuntu#

```

6.

More info about LivenessProbe

```
root@ip-172-31-33-162: /home/ubuntu
IP: 172.17.0.3
Containers:
  liveness:
    Container ID:   docker://59c21c899cad293f00474f85ecdaf8b37ef1e0a97381fa55923287cd7e5863e
    Image:          ubuntu
    Image ID:       docker-pullable://ubuntu@sha256:80dd3c3b9c6cecb9f1667e9290b3bc61b78c2678c02cbdae5f0fea92cc6734ab
    Port:           <none>
    Host Port:      <none>
    Args:
      /bin/sh
      -c
      touch /tmp/healthy; sleep 1000
    State:          Running
      Started:      Mon, 16 Dec 2024 19:49:56 +0000
    Last State:     Terminated
      Reason:       Completed
      Exit Code:    0
      Started:      Mon, 16 Dec 2024 19:33:14 +0000
      Finished:     Mon, 16 Dec 2024 19:49:54 +0000
    Ready:          True
    Restart Count:  2
    Liveness:       exec [cat /tmp/healthy] delay=5s timeout=30s period=5s #success=1 #failure=3
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-9wj22 (ro)
Conditions:
  Type            Status
  Initialized      True
  Ready            True
  ContainersReady  True
  PodScheduled     True
Volumes:
  kube-api-access-9wj22:
    Type:          Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:    kube-root-ca.crt
    ConfigMapOptional: <nil>
```

7.

command: `kubectl describe pod pod-name`

In the description we can see the details about live-ness test command

Some detailed logs info about the LivenessProbe

```
root@ip-172-31-33-162: /home/ubuntu
DownwardAPI: true
QoS Class:   BestEffort
Node-Selectors: <none>
Terminations: node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
               node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type     Reason      Age    From          Message
  ----     -
  Normal   Scheduled   27m    default-scheduler Successfully assigned default/mylivenessprobe to ip-172-31-33-162
  Normal   Pulled      27m    kubelet       Successfully pulled image "ubuntu" in 5.683795099s
  Warning  Unhealthy   24m    kubelet       Liveness probe failed: cat: /tmp/healthy: No such file or d
  Normal   Killing     24m    kubelet       Container liveness failed liveness probe, will be restarted
  Normal   Pulled      24m    kubelet       Successfully pulled image "ubuntu" in 1.838689718s
  Normal   Pulling     7m29s (x3 over 27m) kubelet       Pulling image "ubuntu"
  Normal   Created     7m27s (x3 over 27m) kubelet       Created container liveness
  Normal   Started     7m27s (x3 over 27m) kubelet       Started container liveness
  Normal   Pulled      7m27s    kubelet       Successfully pulled image "ubuntu" in 1.775519435s
root@ip-172-31-33-162: /home/ubuntu#
```

8.

As we can see here Container liveness failed and container was restarted again.